UNIT -2 CLASS ACTIVITY

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1)
Hospital Patient
Admission Analysis

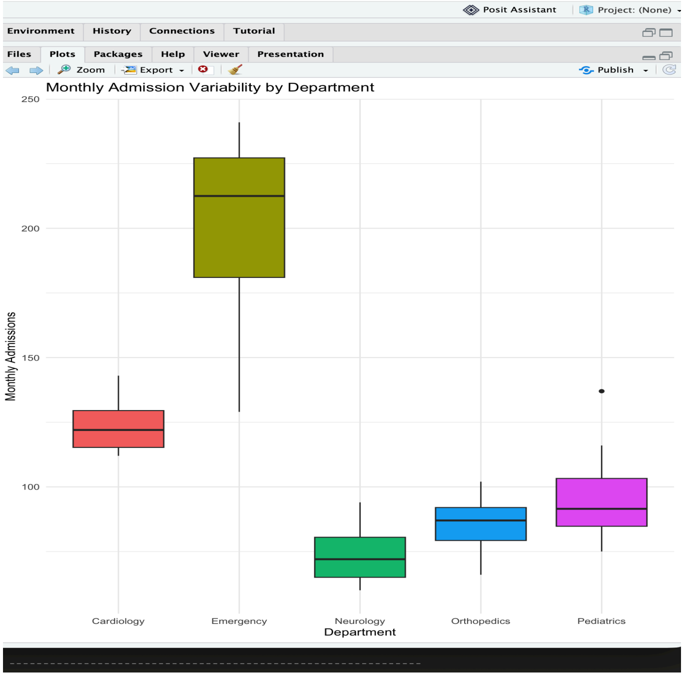
A hospital has collected data on patient admissions across multiple departments (Cardiology, Neurology, Pediatrics, Orthopedics, Emergency) for the past year. Different visualizations (bar chart, stacked bar chart, treemap, box plot, histogram) are available. Evaluate which visualization most effectively communicates department workload, admission variability, and patient distribution. Justify your selection with evidence.

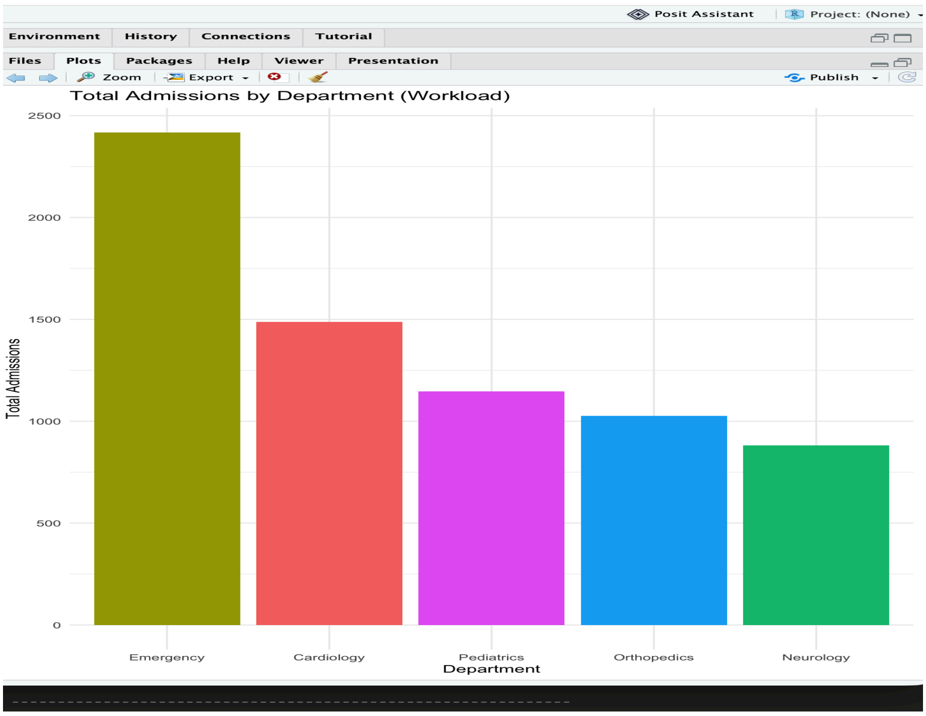
CODE:

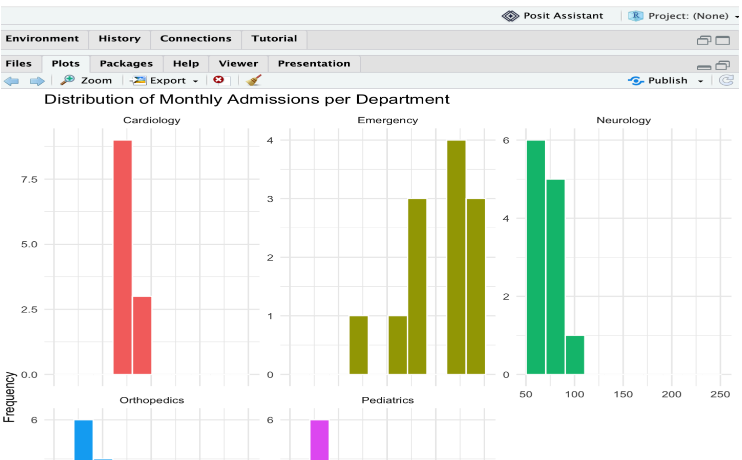
```
library(ggplot2)
library(dplyr)
library(readr)

df <- read_csv("hospital_admissions_1.csv")

glimpse(df)

totals <- df %>%
  group_by(department) %>%
  summarise(total_admissions = sum(admissions)) %>%
  arrange(desc(total_admissions))

p1 <- ggplot(totals, aes(x = reorder(department, -total_admissions),
                         y = total_admissions,
                         fill = department)) +
  geom_col() +
  labs(title = "Total Admissions by Department (Workload)",
       x = "Department", y = "Total Admissions") +
  theme_minimal() +
  theme(legend.position = "none")

print(p1)
```

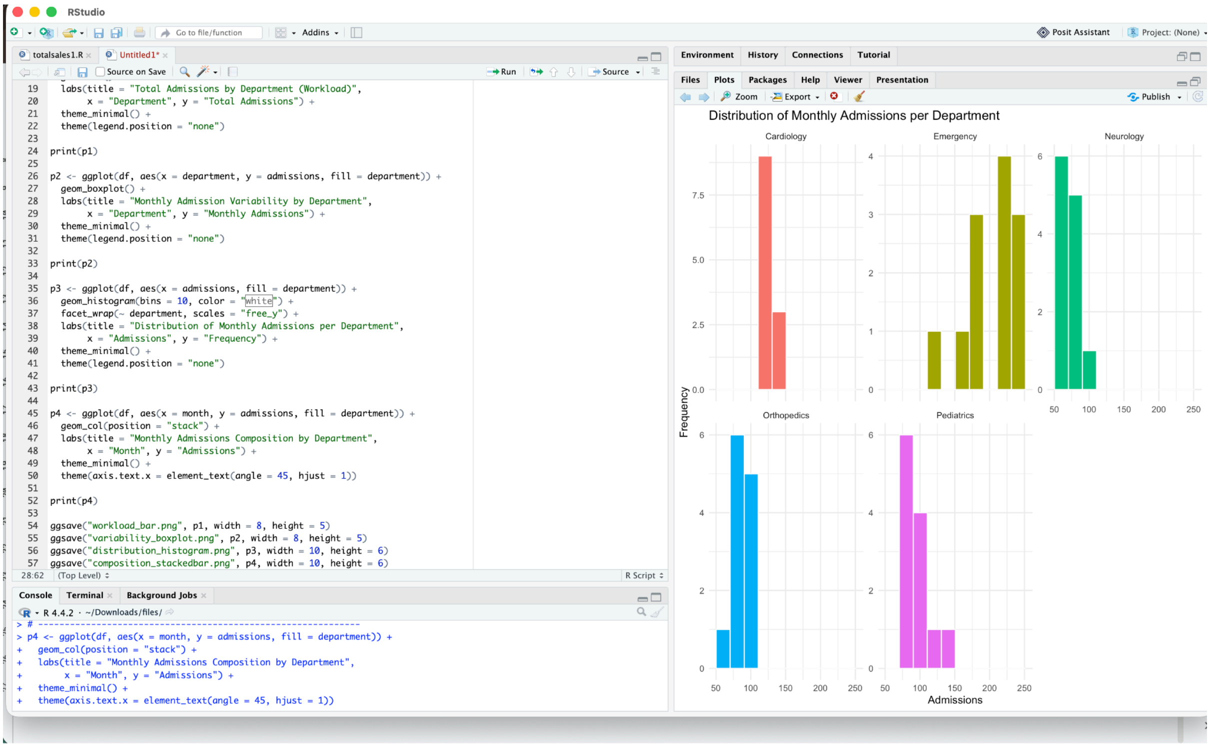
```r
p2 <- ggplot(df, aes(x = department, y = admissions, fill = department)) +
  geom_boxplot() +
  labs(title = "Monthly Admission Variability by Department",
       x = "Department", y = "Monthly Admissions") +
  theme_minimal() +
  theme(legend.position = "none")

print(p2)

p3 <- ggplot(df, aes(x = admissions, fill = department)) +
  geom_histogram(bins = 10, color = "white") +
  facet_wrap(~ department, scales = "free_y") +
  labs(title = "Distribution of Monthly Admissions per Department",
       x = "Admissions", y = "Frequency") +
  theme_minimal() +
  theme(legend.position = "none")

print(p3)

p4 <- ggplot(df, aes(x = month, y = admissions, fill = department)) +
  geom_col(position = "stack") +
  labs(title = "Monthly Admissions Composition by Department",
       x = "Month", y = "Admissions") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

print(p4)

ggsave("workload_bar.png", p1, width = 8, height = 5)
ggsave("variability_boxplot.png", p2, width = 8, height = 5)
ggsave("distribution_histogram.png", p3, width = 10, height = 6)
ggsave("composition_stackedbar.png", p4, width = 10, height = 6)
```

Screenshots:


Posit Assistant
Project: (None)
Environment
History
Connections
Tutorial
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Monthly Admission Variability by Department
Monthly Admissions
250
200
150
100
Cardiology
Emergency
Neurology
Orthopedics
Pediatrics
Department


Posit Assistant
Project: (None)
Environment
History
Connections
Tutorial
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Total Admissions by Department (Workload)
Total Admissions
2500
2000
1500
1000
500
0
Emergency
Cardiology
Pediatrics
Orthopedics
Neurology
Department


Posit Assistant
Project: (None)
Environment
History
Connections
Tutorial
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Distribution of Monthly Admissions per Department
Cardiology
Emergency
Neurology
Orthopedics
Pediatrics
Frequency
7.5
5.0
2.5
0.0
4
3
2
1
0
6
4
2
0
50
100
150
200
250



2)
E-Commerce Customer
Spending Behaviour

An online retailer has customer purchase amounts, order frequency, and product categories. Create multiple visualizations (histogram, violin plot, box plot, density plot, bar chart). Evaluate which visualization best identifies high-value customers, spending variability, and purchasing trends. Recommend the most appropriate visualization for business executives and explain why.

CODE:

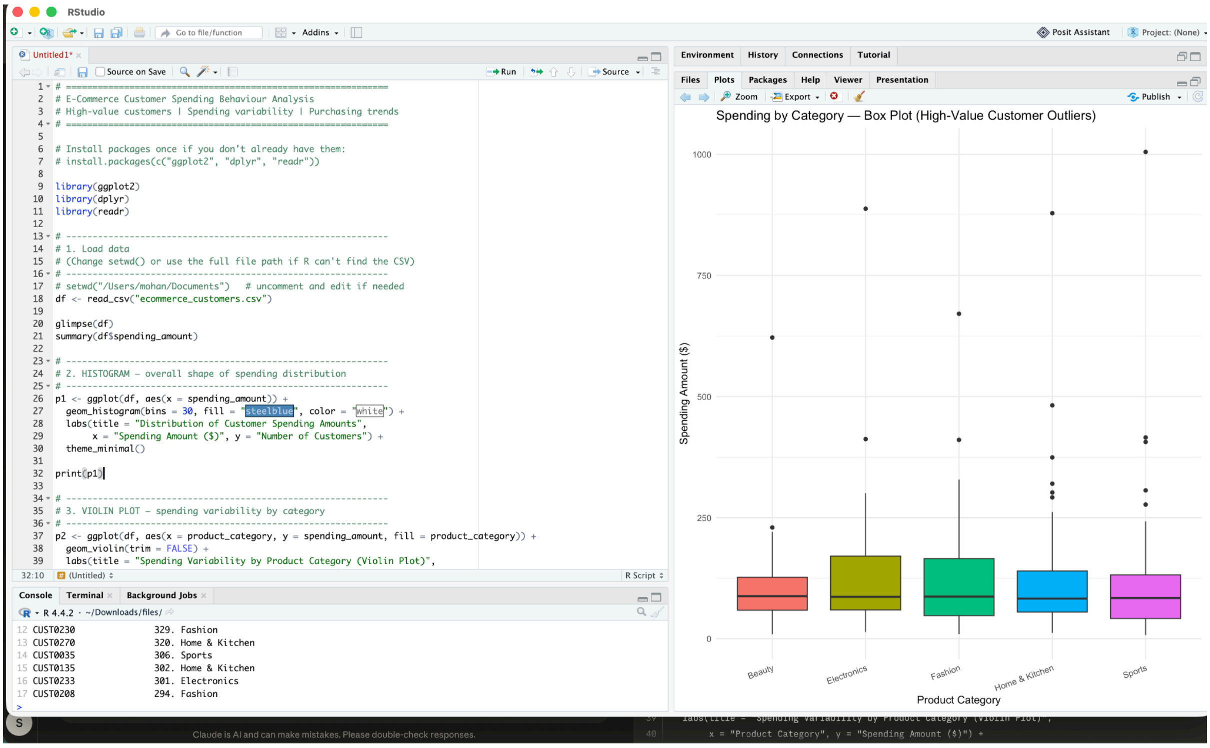
```
# =================================================================
# E-Commerce Customer Spending Behaviour Analysis
# High-value customers | Spending variability | Purchasing trends
# =================================================================

# Install packages once if you don't already have them:
# install.packages(c("ggplot2", "dplyr", "readr"))

library(ggplot2)
library(dplyr)
library(readr)
```

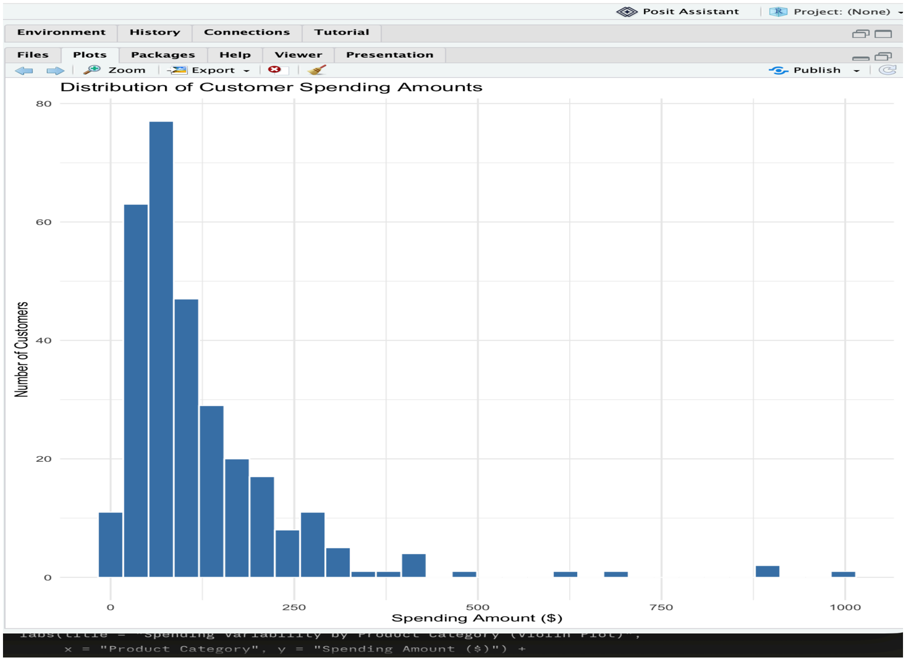
```
# -----------------------------------------------------------
# 1. Load data
# (Change setwd() or use the full file path if R can't find the CSV)
# -----------------------------------------------------------
# setwd("/Users/mohan/Documents")   # uncomment and edit if needed
df <- read_csv("ecommerce_customers.csv")

glimpse(df)
summary(df$spending_amount)

# -----------------------------------------------------------
# 2. HISTOGRAM — overall shape of spending distribution
# -----------------------------------------------------------
p1 <- ggplot(df, aes(x = spending_amount)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  labs(title = "Distribution of Customer Spending Amounts",
       x = "Spending Amount ($)", y = "Number of Customers") +
  theme_minimal()

print(p1)

# -----------------------------------------------------------
# 3. VIOLIN PLOT — spending variability by category
# -----------------------------------------------------------
p2 <- ggplot(df, aes(x = product_category, y = spending_amount, fill = product_category)) +
  geom_violin(trim = FALSE) +
  labs(title = "Spending Variability by Product Category (Violin Plot)",
       x = "Product Category", y = "Spending Amount ($)") +
  theme_minimal() +
  theme(legend.position = "none", axis.text.x = element_text(angle = 20, hjust = 1))

print(p2)

# -----------------------------------------------------------
# 4. BOX PLOT — identify high-value customers (outliers)
# -----------------------------------------------------------
p3 <- ggplot(df, aes(x = product_category, y = spending_amount, fill = product_category)) +
  geom_boxplot() +
  labs(title = "Spending by Category — Box Plot (High-Value Customer Outliers)",
       x = "Product Category", y = "Spending Amount ($)") +
  theme_minimal() +
  theme(legend.position = "none", axis.text.x = element_text(angle = 20, hjust = 1))
```

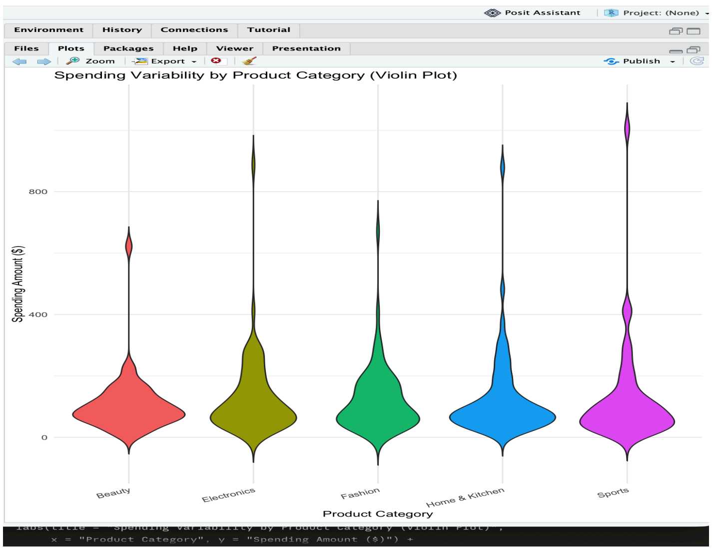
```
print(p3)

# ------------------------------------------------------------
# 5. DENSITY PLOT — purchasing trends across categories
# ------------------------------------------------------------
p4 <- ggplot(df, aes(x = spending_amount, fill = product_category)) +
  geom_density(alpha = 0.4) +
  labs(title = "Spending Density by Product Category",
       x = "Spending Amount ($)", y = "Density") +
  theme_minimal()

print(p4)

# ------------------------------------------------------------
# 6. BAR CHART — total revenue by category (executive view)
# ------------------------------------------------------------
revenue_by_category <- df %>%
  group_by(product_category) %>%
  summarise(total_revenue = sum(spending_amount)) %>%
  arrange(desc(total_revenue))

p5 <- ggplot(revenue_by_category, aes(x = reorder(product_category, -total_revenue),
                                      y = total_revenue,
                                      fill = product_category)) +
  geom_col() +
  labs(title = "Total Revenue by Product Category",
       x = "Product Category", y = "Total Spending ($)") +
  theme_minimal() +
  theme(legend.position = "none")

print(p5)

# ------------------------------------------------------------
# 7. Identify high-value customers using IQR outlier rule
# ------------------------------------------------------------
Q1 <- quantile(df$spending_amount, 0.25)
Q3 <- quantile(df$spending_amount, 0.75)
IQR_val <- Q3 - Q1
upper_bound <- Q3 + 1.5 * IQR_val

high_value_customers <- df %>%
  filter(spending_amount > upper_bound) %>%
  arrange(desc(spending_amount))
```

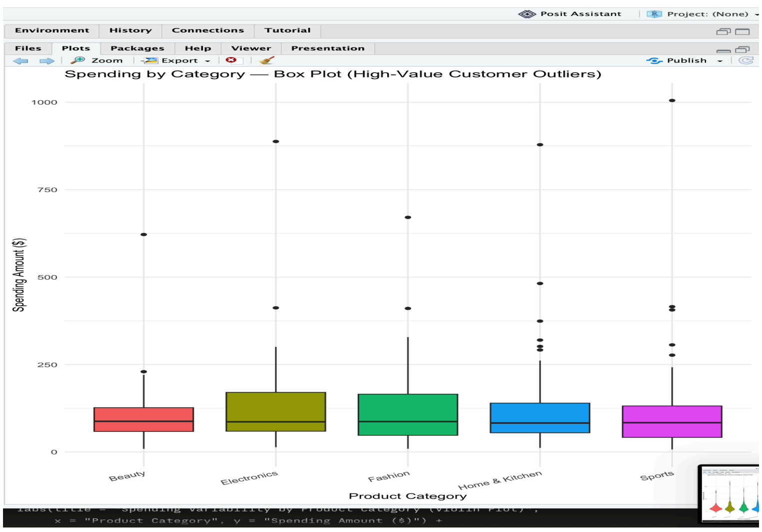
```
cat("High-value threshold (Q3 + 1.5*IQR): $", round(upper_bound, 2), "\n")
cat("Number of high-value customers:", nrow(high_value_customers), "\n")
print(high_value_customers %>% select(customer_id, spending_amount, product_category))

# ------------------------------------------------------------
# 8. Save plots to files (optional)
# ------------------------------------------------------------
ggsave("histogram_spending.png", p1, width = 8, height = 5)
ggsave("violin_spending.png", p2, width = 9, height = 5)
ggsave("boxplot_spending.png", p3, width = 9, height = 5)
ggsave("density_spending.png", p4, width = 8, height = 5)
ggsave("barchart_revenue.png", p5, width = 8, height = 5)
```

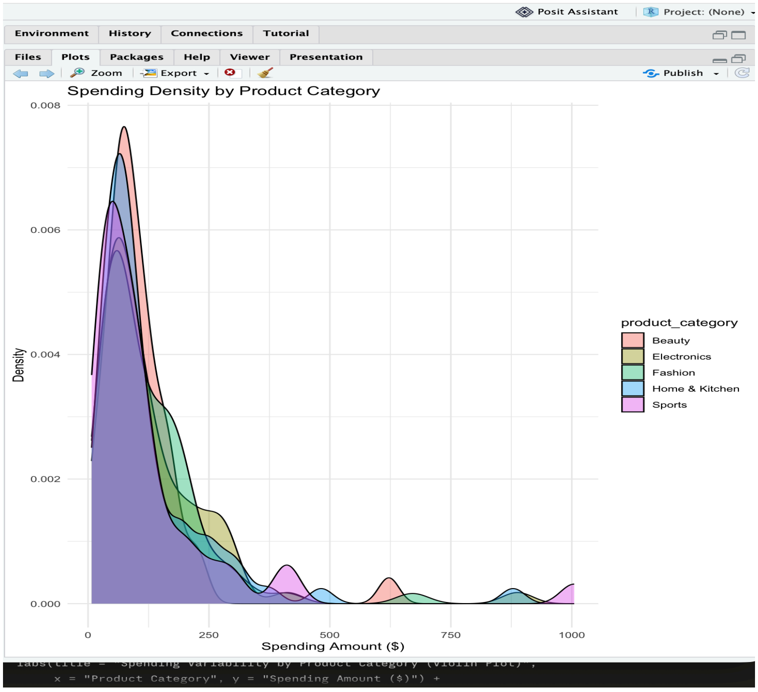
SCREENSHOTS:

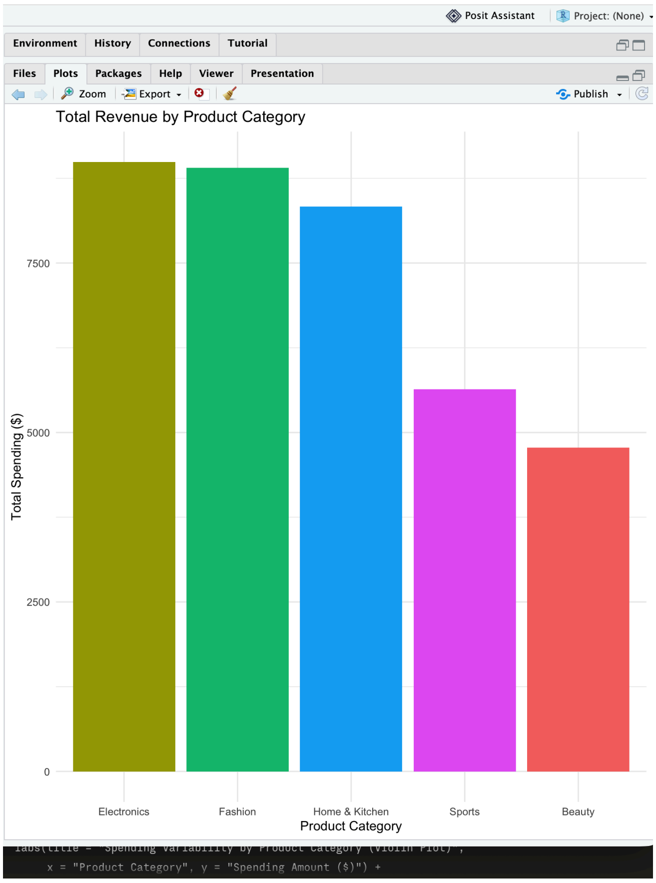



Posit Assistant
Project: (None)
Environment History Connections Tutorial
Files Plots Packages Help Viewer Presentation
Zoom Export Publish
Distribution of Customer Spending Amounts
Number of Customers
0 20 40 60 80
0 250 500 750 1000
Spending Amount ($)
x = "Product Category", y = "Spending Amount ($)") +


Posit Assistant
Project: (None)
Environment History Connections Tutorial
Files Plots Packages Help Viewer Presentation
Zoom Export Publish
Spending Variability by Product Category (Violin Plot)
Spending Amount ($)
0 400 800
Beauty Electronics Fashion Home & Kitchen Sports
Product Category
x = "Product Category", y = "Spending Amount ($)") +


Posit Assistant
Project: (None)
Environment History Connections Tutorial
Files Plots Packages Help Viewer Presentation
Zoom Export Publish
Spending by Category — Box Plot (High-Value Customer Outliers)
Spending Amount ($)
0 250 500 750 1000
Beauty Electronics Fashion Home & Kitchen Sports
Product Category
x = "Product Category", y = "Spending Amount ($)") +


Posit Assistant
Project: (None)
Environment
History
Connections
Tutorial
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Spending Density by Product Category
0.008
0.006
0.004
0.002
0.000
Density
0
250
500
750
1000
Spending Amount ($)
product_category
Beauty
Electronics
Fashion
Home & Kitchen
Sports
x = "Product Category", y = "Spending Amount ($)") +


Posit Assistant
Project: (None)
Environment
History
Connections
Tutorial
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Total Revenue by Product Category
7500
5000
2500
0
Total Spending ($)
Electronics
Fashion
Home & Kitchen
Sports
Beauty
Product Category
x = "Product Category", y = "Spending Amount ($)") +

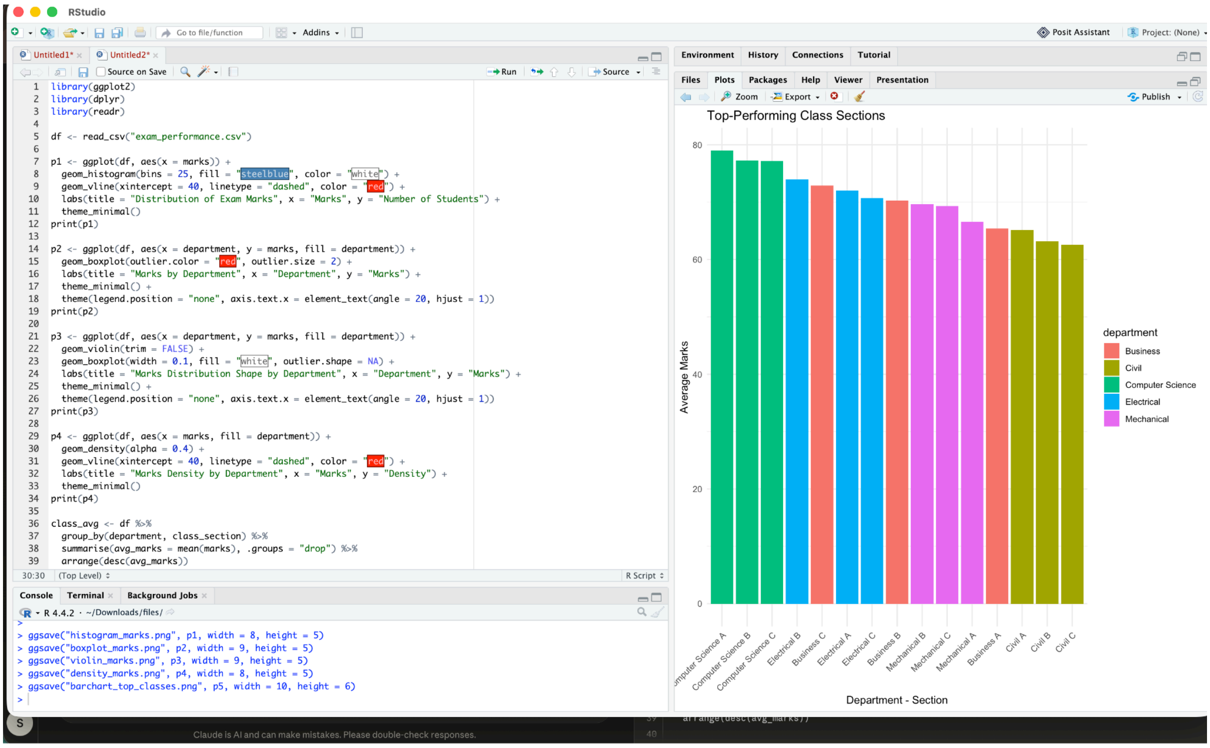
3)
University Examination
Performance Dashboard

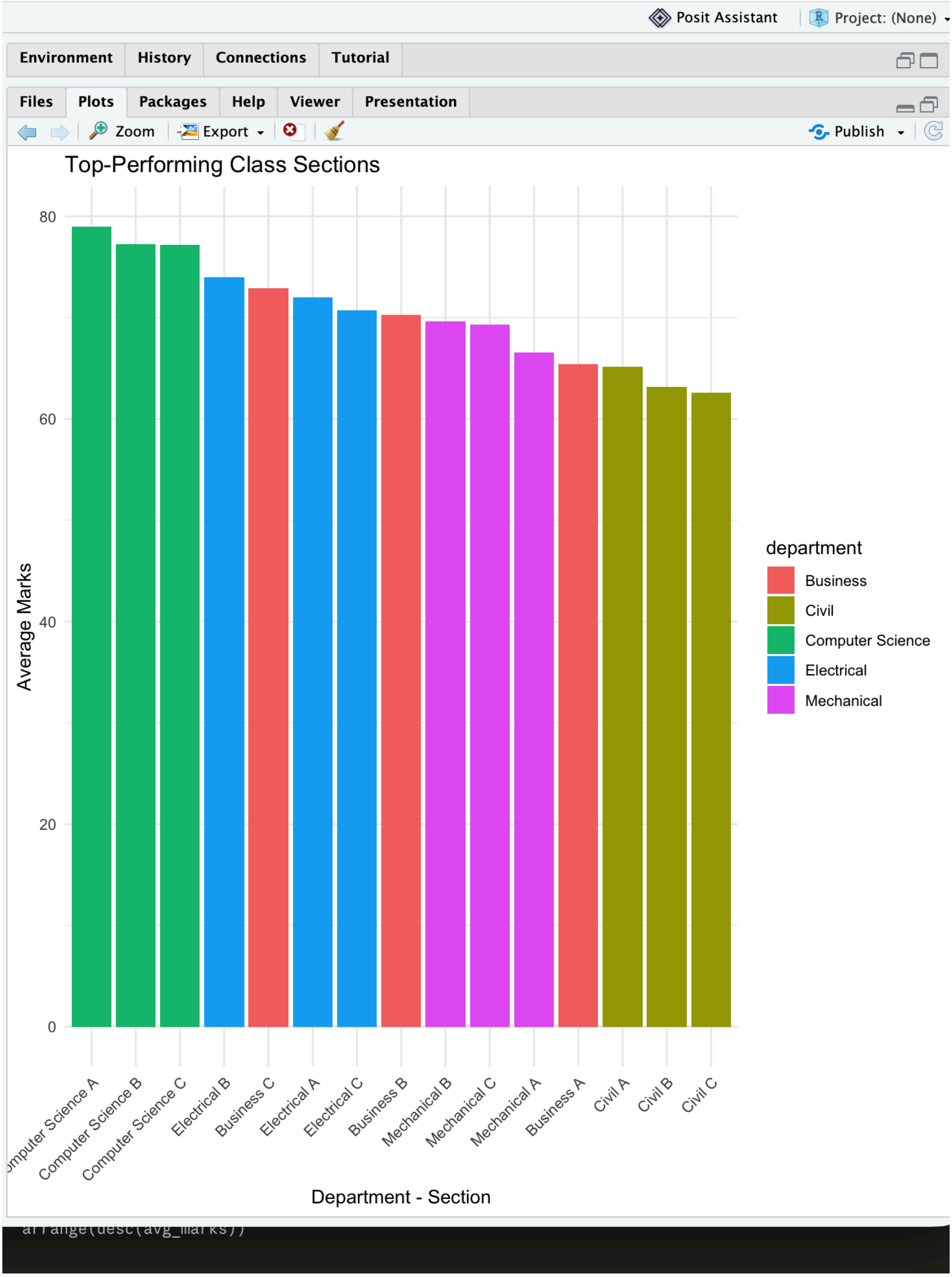
A university wants to compare student marks across departments and identify score distributions, outliers, and top-performing classes. Generate different visualizations and evaluate which charts provide the clearest insights for academic decision-making. Recommend improvements to the visualization dashboard.

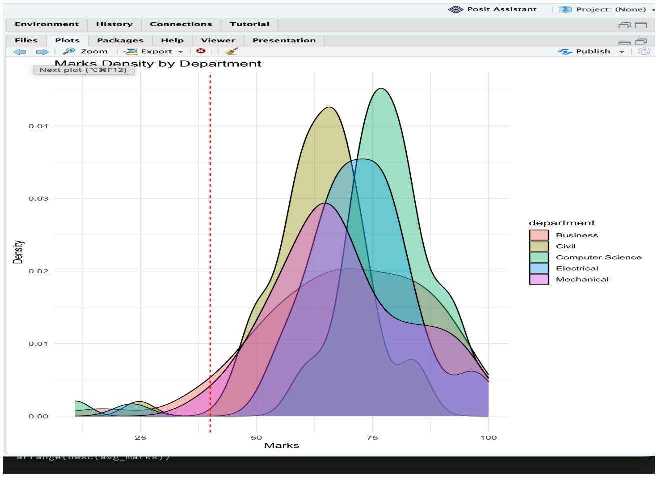
CODE:
```
library(ggplot2)
library(dplyr)
library(readr)

df <- read_csv("exam_performance.csv")

p1 <- ggplot(df, aes(x = marks)) +
  geom_histogram(bins = 25, fill = "steelblue", color = "white") +
  geom_vline(xintercept = 40, linetype = "dashed", color = "red") +
  labs(title = "Distribution of Exam Marks", x = "Marks", y = "Number of Students") +
  theme_minimal()
print(p1)

p2 <- ggplot(df, aes(x = department, y = marks, fill = department)) +
  geom_boxplot(outlier.color = "red", outlier.size = 2) +
  labs(title = "Marks by Department", x = "Department", y = "Marks") +
  theme_minimal() +
  theme(legend.position = "none", axis.text.x = element_text(angle = 20, hjust = 1))
print(p2)

p3 <- ggplot(df, aes(x = department, y = marks, fill = department)) +
  geom_violin(trim = FALSE) +
  geom_boxplot(width = 0.1, fill = "white", outlier.shape = NA) +
  labs(title = "Marks Distribution Shape by Department", x = "Department", y = "Marks") +
  theme_minimal() +
  theme(legend.position = "none", axis.text.x = element_text(angle = 20, hjust = 1))
print(p3)

p4 <- ggplot(df, aes(x = marks, fill = department)) +
  geom_density(alpha = 0.4) +
  geom_vline(xintercept = 40, linetype = "dashed", color = "red") +
  labs(title = "Marks Density by Department", x = "Marks", y = "Density") +
  theme_minimal()
print(p4)
```

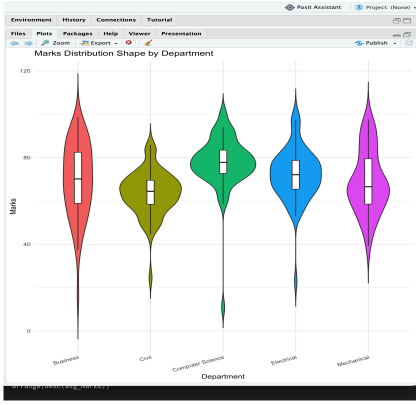
```
class_avg <- df %>%
  group_by(department, class_section) %>%
  summarise(avg_marks = mean(marks), .groups = "drop") %>%
  arrange(desc(avg_marks))

p5 <- ggplot(class_avg, aes(x = reorder(paste(department, class_section), -avg_marks),
                            y = avg_marks, fill = department)) +
  geom_col() +
  labs(title = "Top-Performing Class Sections", x = "Department - Section", y = "Average
Marks") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
print(p5)

outliers <- df %>%
  group_by(department) %>%
  mutate(
    Q1 = quantile(marks, 0.25),
    Q3 = quantile(marks, 0.75),
    IQR_val = Q3 - Q1,
    lower_bound = Q1 - 1.5 * IQR_val,
    upper_bound = Q3 + 1.5 * IQR_val
  ) %>%
  filter(marks < lower_bound | marks > upper_bound) %>%
  select(student_id, department, class_section, marks) %>%
  arrange(department, desc(marks))
print(outliers)

pass_summary <- df %>%
  group_by(department) %>%
  summarise(
    total_students = n(),
    pass_count = sum(marks >= 40),
    pass_rate_pct = round(100 * pass_count / total_students, 1)
  ) %>%
  arrange(desc(pass_rate_pct))
print(pass_summary)

ggsave("histogram_marks.png", p1, width = 8, height = 5)
ggsave("boxplot_marks.png", p2, width = 9, height = 5)
ggsave("violin_marks.png", p3, width = 9, height = 5)
ggsave("density_marks.png", p4, width = 8, height = 5)
ggsave("barchart_top_classes.png", p5, width = 10, height = 6)
```

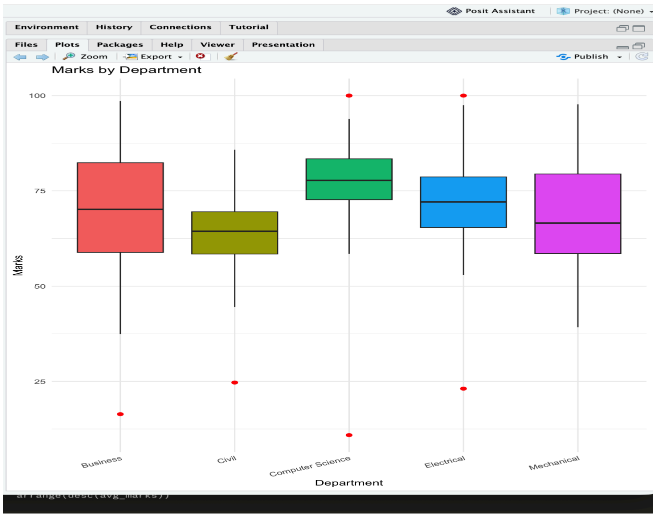
SCREENSHOTS:

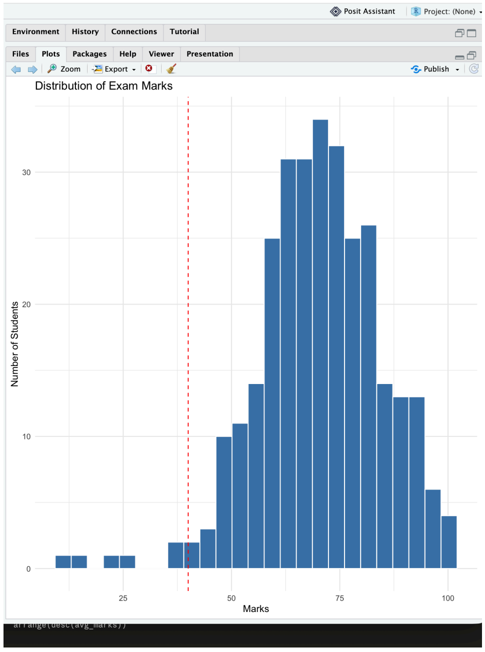



Posit Assistant
Project: (None)
Environment
History
Connections
Tutorial
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Top-Performing Class Sections
80
60
40
20
0
Average Marks
Computer Science B
Computer Science C
Electrical B
Business C
Electrical A
Electrical C
Business B
Mechanical B
Mechanical C
Mechanical A
Business A
Civil A
Civil B
Civil C
Department - Section
department
Business
Civil
Computer Science
Electrical
Mechanical
arrange(desc(avg_marks))


Marks Density by Department
Density
Marks
department
Business
Civil
Computer Science
Electrical
Mechanical


Marks Distribution Shape by Department
Marks
Department
Business
Civil
Computer Science
Electrical
Mechanical


Marks by Department
Marks
Department
Business
Civil
Computer Science
Electrical
Mechanical


Distribution of Exam Marks
Number of Students
Marks

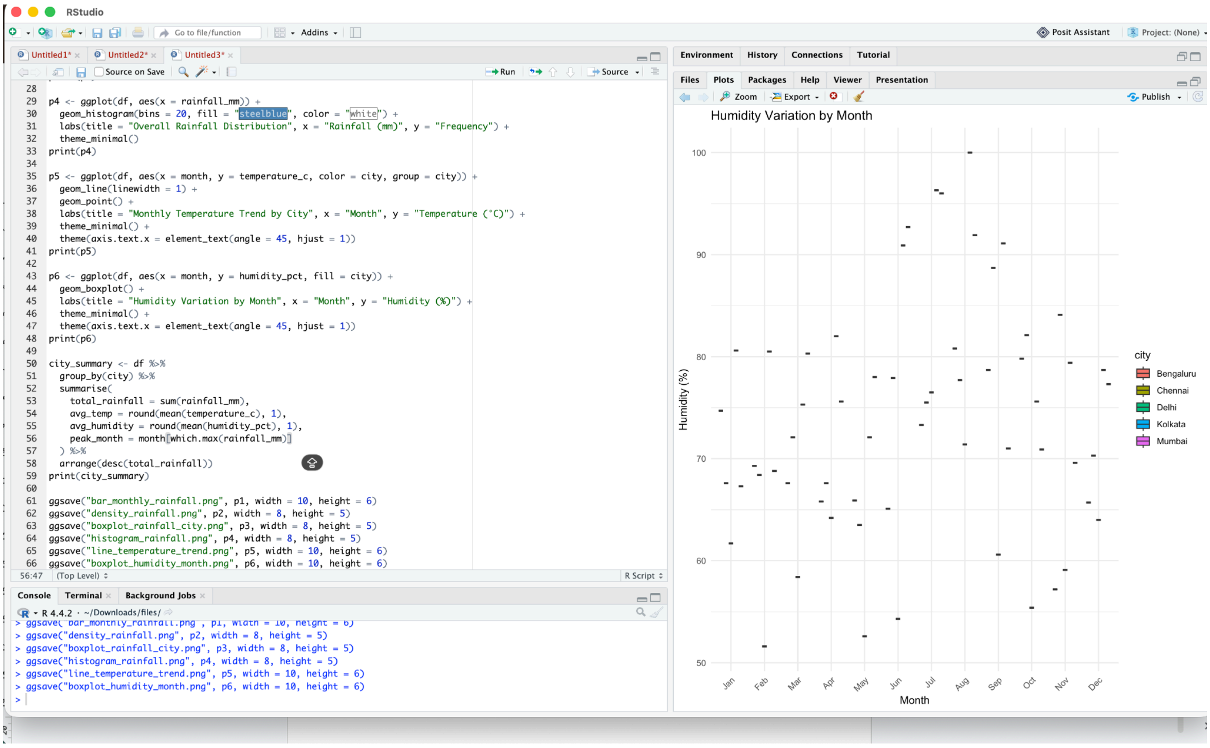
4)Climate and Rainfall
Distribution Study

Meteorological data contains monthly rainfall, temperature, and humidity for multiple cities. Produce histograms, density plots, box plots, and bar charts. Evaluate which visualization best represents seasonal rainfall variation and climate distribution. Compare at least three visualizations and justify the most suitable one for environmental

CODE:
```
library(ggplot2)
library(dplyr)
library(readr)

df <- read_csv("climate_data.csv")
df$month <- factor(df$month, levels = c("Jan","Feb","Mar","Apr","May","Jun",
                                        "Jul","Aug","Sep","Oct","Nov","Dec"))

p1 <- ggplot(df, aes(x = month, y = rainfall_mm, fill = city)) +
  geom_col(position = "dodge") +
  labs(title = "Monthly Rainfall by City", x = "Month", y = "Rainfall (mm)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
print(p1)

p2 <- ggplot(df, aes(x = rainfall_mm, fill = city)) +
  geom_density(alpha = 0.4) +
  labs(title = "Rainfall Distribution by City", x = "Rainfall (mm)", y = "Density") +
  theme_minimal()
print(p2)

p3 <- ggplot(df, aes(x = city, y = rainfall_mm, fill = city)) +
  geom_boxplot(outlier.color = "red", outlier.size = 2) +
  labs(title = "Rainfall Spread by City (Outliers Highlighted)", x = "City", y = "Rainfall
(mm)") +
  theme_minimal() +
  theme(legend.position = "none")
print(p3)

p4 <- ggplot(df, aes(x = rainfall_mm)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "white") +
  labs(title = "Overall Rainfall Distribution", x = "Rainfall (mm)", y = "Frequency") +
  theme_minimal()
print(p4)
```

```r
p5 <- ggplot(df, aes(x = month, y = temperature_c, color = city, group = city)) +
  geom_line(linewidth = 1) +
  geom_point() +
  labs(title = "Monthly Temperature Trend by City", x = "Month", y = "Temperature (°C)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
print(p5)

p6 <- ggplot(df, aes(x = month, y = humidity_pct, fill = city)) +
  geom_boxplot() +
  labs(title = "Humidity Variation by Month", x = "Month", y = "Humidity (%)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
print(p6)

city_summary <- df %>%
  group_by(city) %>%
  summarise(
    total_rainfall = sum(rainfall_mm),
    avg_temp = round(mean(temperature_c), 1),
    avg_humidity = round(mean(humidity_pct), 1),
    peak_month = month[which.max(rainfall_mm)]
  ) %>%
  arrange(desc(total_rainfall))
print(city_summary)

ggsave("bar_monthly_rainfall.png", p1, width = 10, height = 6)
ggsave("density_rainfall.png", p2, width = 8, height = 5)
ggsave("boxplot_rainfall_city.png", p3, width = 8, height = 5)
ggsave("histogram_rainfall.png", p4, width = 8, height = 5)
ggsave("line_temperature_trend.png", p5, width = 10, height = 6)
ggsave("boxplot_humidity_month.png", p6, width = 10, height = 6)
```

SCREENSHOTS:

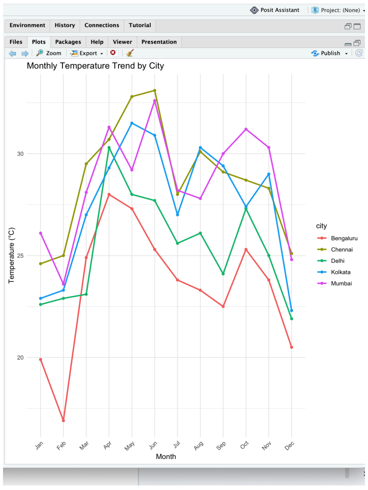

RStudio
Go to file/function
Addins
Untitled1*
Untitled2*
Untitled3*
Source on Save
Run
Source
p4 <- ggplot(df, aes(x = rainfall_mm)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "white") +
  labs(title = "Overall Rainfall Distribution", x = "Rainfall (mm)", y = "Frequency") +
  theme_minimal()
print(p4)

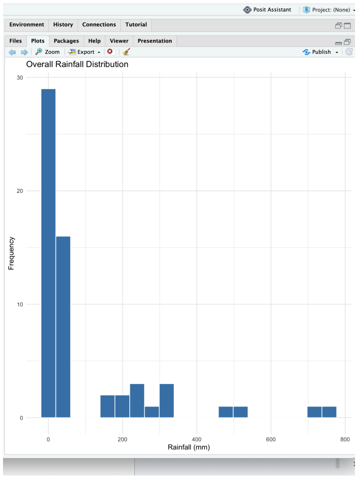
p5 <- ggplot(df, aes(x = month, y = temperature_c, color = city, group = city)) +
  geom_line(linewidth = 1) +
  geom_point() +
  labs(title = "Monthly Temperature Trend by City", x = "Month", y = "Temperature (°C)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
print(p5)

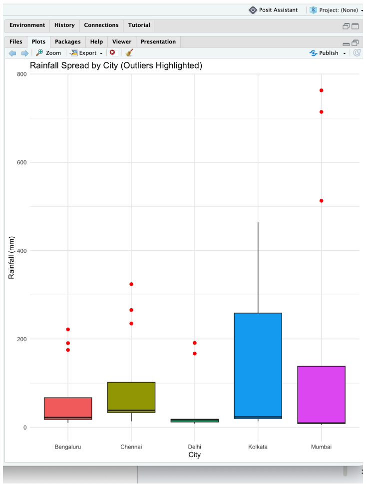
p6 <- ggplot(df, aes(x = month, y = humidity_pct, fill = city)) +
  geom_boxplot() +
  labs(title = "Humidity Variation by Month", x = "Month", y = "Humidity (%)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
print(p6)

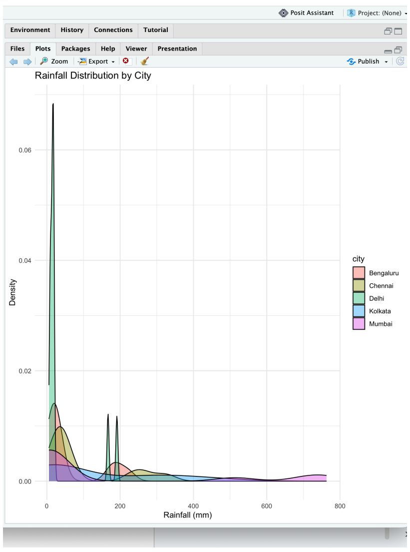
city_summary <- df %>%
  group_by(city) %>%
  summarise(
    total_rainfall = sum(rainfall_mm),
    avg_temp = round(mean(temperature_c), 1),
    avg_humidity = round(mean(humidity_pct), 1),
    peak_month = month[which.max(rainfall_mm)]
) %>%
  arrange(desc(total_rainfall))
print(city_summary)

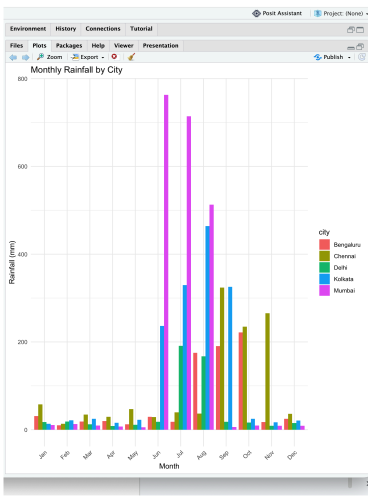
ggsave("bar_monthly_rainfall.png", p1, width = 10, height = 6)
ggsave("density_rainfall.png", p2, width = 8, height = 5)
ggsave("boxplot_rainfall_city.png", p3, width = 8, height = 5)
ggsave("histogram_rainfall.png", p4, width = 8, height = 5)
ggsave("line_temperature_trend.png", p5, width = 10, height = 6)
ggsave("boxplot_humidity_month.png", p6, width = 10, height = 6)
56:47
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.4.2 · ~/Downloads/files/
> ggsave("density_rainfall.png", p2, width = 8, height = 5)
> ggsave("boxplot_rainfall_city.png", p3, width = 8, height = 5)
> ggsave("histogram_rainfall.png", p4, width = 8, height = 5)
> ggsave("line_temperature_trend.png", p5, width = 10, height = 6)
> ggsave("boxplot_humidity_month.png", p6, width = 10, height = 6)
>
Posit Assistant
Project: (None)
Environment
History
Connections
Tutorial
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Humidity Variation by Month
Humidity (%)
100
90
80
70
60
50
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Month
city
Bengaluru
Chennai
Delhi
Kolkata
Mumbai


Monthly Temperature Trend by City
Temperature (°C)
30
25
20
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Month
city
Bengaluru
Chennai
Delhi
Kolkata
Mumbai


Overall Rainfall Distribution
Frequency
30
20
10
0
0
200
400
600
800
Rainfall (mm)


Rainfall Spread by City (Outliers Highlighted)
Rainfall (mm)
800
600
400
200
0
Bengaluru
Chennai
Delhi
Kolkata
Mumbai
City


Rainfall Distribution by City
Density
0.06
0.04
0.02
0.00
0
200
400
600
800
Rainfall (mm)
city
Bengaluru
Chennai
Delhi
Kolkata
Mumbai


Monthly Rainfall by City
Rainfall (mm)
800
600
400
200
0
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Month
city
Bengaluru
Chennai
Delhi
Kolkata
Mumbai

5)
Bank Loan Risk
Assessment

A financial institution has customer loan amounts, repayment history, income levels, and credit scores. Develop visualizations showing loan distributions and customer segments. Evaluate which visualization best supports loan approval decisions by highlighting risk patterns, skewness, and outliers. Recommend the visualization that should be included in a management dashboard.

CODE:
```
library(ggplot2)
library(dplyr)
library(readr)

df <- read_csv("loan_risk.csv")
df$repayment_history <- factor(df$repayment_history, levels = c("On-time", "Late",
"Default"))

p1 <- ggplot(df, aes(x = loan_amount)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  labs(title = "Distribution of Loan Amounts", x = "Loan Amount ($)", y = "Number of
Customers") +
  theme_minimal()
print(p1)

p2 <- ggplot(df, aes(x = loan_amount, fill = repayment_history)) +
  geom_density(alpha = 0.4) +
  labs(title = "Loan Amount Density by Repayment History", x = "Loan Amount ($)", y =
"Density") +
  theme_minimal()
print(p2)

p3 <- ggplot(df, aes(x = repayment_history, y = loan_amount, fill = repayment_history)) +
  geom_boxplot(outlier.color = "red", outlier.size = 2) +
  labs(title = "Loan Amount by Repayment History (Outliers Highlighted)",
       x = "Repayment History", y = "Loan Amount ($)") +
  theme_minimal() +
  theme(legend.position = "none")
print(p3)

df$credit_band <- cut(df$credit_score,
                      breaks = c(300, 579, 669, 739, 850),
                      labels = c("Poor", "Fair", "Good", "Excellent"))
```

```r
default_rate <- df %>%
  group_by(credit_band) %>%
  summarise(
    total_customers = n(),
    defaults = sum(repayment_history == "Default"),
    default_rate_pct = round(100 * defaults / total_customers, 1)
  )

p4 <- ggplot(default_rate, aes(x = credit_band, y = default_rate_pct, fill = credit_band)) +
  geom_col() +
  labs(title = "Default Rate by Credit Band", x = "Credit Band", y = "Default Rate (%)") +
  theme_minimal() +
  theme(legend.position = "none")
print(p4)

p5 <- ggplot(df, aes(x = credit_score, y = loan_amount, color = repayment_history)) +
  geom_point(alpha = 0.6) +
  labs(title = "Credit Score vs Loan Amount by Repayment Status",
       x = "Credit Score", y = "Loan Amount ($)") +
  theme_minimal()
print(p5)

outliers <- df %>%
  group_by(repayment_history) %>%
  mutate(
    Q1 = quantile(loan_amount, 0.25),
    Q3 = quantile(loan_amount, 0.75),
    IQR_val = Q3 - Q1,
    upper_bound = Q3 + 1.5 * IQR_val
  ) %>%
  filter(loan_amount > upper_bound) %>%
  select(customer_id, credit_score, loan_amount, repayment_history) %>%
  arrange(desc(loan_amount))
print(outliers)

print(default_rate)
ggsave("histogram_loan_amount.png", p1, width = 8, height = 5)
ggsave("density_loan_repayment.png", p2, width = 8, height = 5)
ggsave("boxplot_loan_repayment.png", p3, width = 8, height = 5)
ggsave("barchart_default_rate.png", p4, width = 8, height = 5)
ggsave("scatter_credit_vs_loan.png", p5, width = 9, height = 6)
```

SCREENSHOTS: